SCRIBE: Scoring Generated Text Against Human Text With Counts Alone

Odin Loch

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Code and data recipes: the Cypha repository, github.com/odin-loki/Cypha (`native/include/cypha/eval/scribe.hpp`, `native/tools/scribe.cpp`, `scripts/scribe_eval/`)

Abstract

SCRIBE is a metric for generated text that uses no neural network. It computes about twenty statistics of a text from counts. They cover how much each word's context predicts it beyond its frequency, for words and for word classes; whether the words are real; whether sentences are well formed; how much the sentences share vocabulary and topic, using count-based word vectors; repetition; copying of the training text; and overlap with a prompt. Each statistic is read as a robust z-score against human reference text of the same genre and length. Lexical and syntactic failures count as categorical, saturating evidence, coherence as graded evidence, and repetition only beyond what 99% of human samples show. SCRIBE v2 was designed on development text and then frozen. On fresh Wikipedia text it is the only metric tested that passes all five pairs of a control ladder: human text, then sentences shuffled across documents, words shuffled within sentences, random words, random letters and a repeated phrase, with pairwise accuracies of 0.975 to 1.000. GPT-2 perplexity, a CoLA acceptability classifier, embedding coherence and the usual diversity statistics each pass one or two pairs. SCRIBE v2 also passes on web text, and 3 of 5 pairs on novels. It follows seven of nine held-out kinds of damage and catches copying, prompt echo, keyword stuffing and loops. In its prompted form it separates right from wrong answers to Dolly questions (accuracy 0.961). On HANNA stories it correlates with human coherence ratings (Kendall 0.194) better than BLEU and ROUGE-L and as well as ChatGPT, though there a plain repetition count does as well. It is stable across seeds, across training-text sizes from 8 to 128 MiB, and with 100 or more reference samples per length bucket. It scores about 1,000 texts per second on one CPU core with 0.4 GB of memory. Its blind spots are sentence order, truncated sentences, factual errors, and text that is more predictable than human text: it rates GPT-2 top-k samples above human web text, and it cannot rank models that all write word salad. Of twenty predictions registered before testing, eight hit, eight missed and four could not be tested.

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1. Introduction

Most measures of a language model say how well it predicts human text: perplexity, or bits per character. They do not say whether the text the model writes is any good. A model can compress text well and still write word salad. The usual ways to rate generated text use a second large neural network: perplexity under another language model, similarity in an embedding space, MAUVE, a trained classifier, or a language model asked for a rating. These need a downloaded model and, to be fast, a GPU. They also grade a model with machinery of the same kind as the model under test, with its own blind spots and biases.

This paper asks how far counts alone can go. A count-based metric is cheap and transparent, and it can be rebuilt for a new genre from a few megabytes of human text. It is also easy to get wrong. A simple surface score `S = H·D·L` (the share of words in a vocabulary, times a diversity term, times a loop penalty) shows how. On fresh Wikipedia text it rates real passages with their words shuffled (0.925) and random dictionary words (0.895) about as high as the real passages (0.915). A metric that cannot tell word salad from prose cannot judge a language model.

This paper describes SCRIBE, the metric built to replace it, and the tests it had to pass before it was allowed to judge anything. SCRIBE computes about twenty statistics of a text from counts: n-gram probabilities, word

classes, word co-occurrence vectors, repetition and overlap. It converts each statistic into a signed distance from human text of the same genre and length, read from a sample of human reference texts, and combines the distances into one deviance with a fixed rule. The score is high when a text is typical of human text on every statistic, and falls with the evidence that it is not.

The design rule of this work is that a metric must pass control ladders before it is used. A control ladder is a sequence of texts whose order nobody disputes: real text, then the same text with its sentences shuffled across documents, then with its words shuffled inside sentences, then random dictionary words, then random letters, then one phrase repeated. A metric passes a rung pair when it orders at least 95% of all (upper, lower) pairs correctly and the confidence intervals of the rung means do not overlap. Two earlier versions of SCRIBE failed this test: on the fresh ladder of this paper, v1 passes 2 of 5 pairs and the surface score 1 of 5. This paper presents v2, which was designed on two development ladders and then frozen, and which passes all five pairs on fresh text it had never seen.

Passing a ladder is necessary, not sufficient. We therefore test SCRIBE v2, and standard metrics next to it, on: held-out degradations it was not designed on; language models whose quality order is known (n-gram orders, and GPT-2 sizes under two decoding methods); three published sets of human judgments (the MAUVE human evaluation, HANNA and OpenMEVA); question answering; adversarial attacks; stability with length, seed, genre, reference size and training-text size; ablations of every part; and cost. Twenty predictions were registered before the first v2 run. Each is scored hit or miss, and four could not be tested.

Contributions.

1. A complete, neural-network-free metric for generated text, SCRIBE v2, defined by human-relative z-scores and a fixed combination rule, with a header-only C++ implementation that scores about a thousand texts per second on one CPU core.
2. Evidence that a per-word context gain, the log-probability of each word given its context minus its log-probability from frequency alone, read against human text and computed for both words and word classes, separates word salad from rare-topic human text far better than raw n-gram probability (Section 4). The quantity itself is the per-word form of SLOR (Pauls and Klein, 2012; Lau et al., 2017), known as a fluency measure.
3. A design history that keeps the failures: v0 and v1, every combination that did not pass, and why the final structure, with categorical form failures and graded coherence, was needed.
4. A broad meta-evaluation of SCRIBE and of standard metrics on the same texts, with pre-registered predictions, bootstrap intervals, and the cases where SCRIBE fails.

2. Related work

Reference-based metrics. BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), METEOR (Banerjee and Lavie, 2005) and chrF (Popović, 2015) count n-gram overlap with a reference. BERTScore (Zhang et al., 2020), MoverScore (Zhao et al., 2019), BLEURT (Sellam et al., 2020) and BARTScore (Yuan et al., 2021) compare texts through neural representations. All of them need a reference written for the same input, and their validity for open-ended generation, where many different texts are good, is weak (Novikova et al., 2017; Reiter, 2018; Gehrmann et al., 2023).

Reference-free and distributional metrics. Perplexity under a large language model rewards predictable text and is known to prefer degenerate, repetitive output (Holtzman et al., 2020). Diversity statistics such as distinct-n (Li et al., 2016), self-BLEU (Zhu et al., 2018), repetition rates (Welleck et al., 2020) and the Zipf coefficient (Holtzman et al., 2020) each capture one property. MAUVE (Pillutla et al., 2021) compares the distribution of model text with human text in the embedding space of a large language model and correlates well with human judgments of model families. Meister and Cotterell (2021) compare model text and human text on statistical tendencies such as type–token ratios and length distributions; SCRIBE shares their premise that human text sets the norm, but scores individual texts. HUSE (Hashimoto et al., 2019) combines human judgments with model

probabilities to estimate how distinguishable model text is from human text. GRUEN (Zhu and Bhat, 2020) and UNION (Guan and Huang, 2020) learn quality from perturbations with neural encoders.

Fluency from language-model probability. SLOR, the syntactic log-odds ratio, subtracts a sentence's unigram log-probability from its language-model log-probability and divides by its length, so that rare words do not count as disfluency (Pauls and Klein, 2012). It predicts human acceptability judgments (Lau et al., 2017) and, with a neural language model, rates sentence fluency without a reference (Kann et al., 2018). SCRIBE's context gains s3g and s2g are SLOR computed per word with a count trigram, for words and for word classes; what SCRIBE adds is reading them against a human reference distribution and combining them with the other axes.

Language models as judges. Prompting a large language model for a rating (G-Eval, Liu et al., 2023; Zheng et al., 2023) gives strong correlations with people, at a large cost and with the model's own biases; HANNA (Chhun et al., 2022, 2024) measures this for story generation.

Coherence. The entity grid (Barzilay and Lapata, 2008) models how entities move between sentences; sentence-order and cohesion models (Lapata, 2003) score local coherence. SCRIBE's lexical cohesion statistic is a count-only form of the entity-grid idea, and its sentence vectors follow the SIF construction of Arora et al. (2017) on PPMI-SVD word vectors (Levy et al., 2015).

Meta-evaluation. OpenMEVA (Guan et al., 2021) and the perturbation checklists of Sai et al. (2021), following CheckList (Ribeiro et al., 2020), test metrics with controlled perturbations; HANNA and the MAUVE human evaluation supply human ratings. Control ladders are a simple, strict version of perturbation testing: every rung must be ordered, with a pass threshold set in advance.

Detection. Detectors of machine text, such as GLTR (Gehrmann et al., 2019), the GPT-2 output detector (Solaiman et al., 2019) and DetectGPT (Mitchell et al., 2023), exploit the fact that sampled text is more predictable than human text. Ippolito et al. (2020) show that the decoding settings that fool people most are the easiest for detectors. This tension appears in SCRIBE's results (Section 6.3).

3. SCRIBE v2

3.1 Principles

1. **Human text is the norm.** Every statistic is read as a distance from human text of the same genre and length bucket, never as an absolute value. A text that matches human text on every statistic scores near the top.
2. **Independence.** Everything SCRIBE learns comes from human text disjoint from the texts it scores and from the training text of any model it scores. It never uses the model under test.
3. **Counts only.** No neural network is trained or run. The statistics come from n-gram counts, word classes, co-occurrence counts and their factorisation.
4. **No use before passing.** A version is used to judge models only after it passes its control ladder on text it was not designed on.

3.2 The pack and the reference

SCRIBE needs two disjoint sets of human text of the target genre:

- **The pack** trains the statistics: a lexicon (words seen at least twice), an interpolated Kneser–Ney word trigram (Kneser and Ney, 1995; Chen and Goodman, 1999), a skeleton trigram over word classes, sentence-level IDF, unigram frequencies, the set of word 8-grams, and 300-dimensional PPMI-SVD word vectors with SIF sentence embeddings (weight $a/(a + p(w))$, $a = 10^{-3}$, first principal component removed).

- **The reference** supplies the human distribution of every statistic. It is cut into samples by the same rule used for the texts being scored: consecutive whole sentences, 400–600 bytes with at least three sentences (standard bucket), or 60–260 bytes (short bucket). For Wikipedia, the pack is 32 MiB of enwik8 and the reference has 1,909 standard and 1,729 short samples.

3.3 Statistics

Table 1 lists the statistics. “Side” says which direction is bad: + means lower than human is bad, − means higher is bad, ± means either.

Table 1. SCRIBE v2 statistics (all from counts).

Axis	Statistic	Definition	Side
L	s1 lexical validity	share of word tokens that are in the lexicon or numeric	+
S	s2g skeleton gain	mean over tokens of `log p(c_t | c_{t−2}, c_{t−1}) − log p(c_t)` for the class sequence: the 150 most frequent words keep their identity, other words map to one of 9 suffix classes, numbers to one class; trigram and unigram trained on the pack	+
S	s3g context gain	mean over in-lexicon words of `log p_KN(w_t | w_{t−2}, w_{t−1}) − log p(w_t)`: how much the two preceding words predict each word, beyond its frequency	+
S	s4 well-formedness	pass rate of six checks: valid UTF-8, sentence-initial capitals, terminal punctuation, balanced brackets, paired quotes, no cut-off last word	+
C	c0 word cohesion	mean cosine over all pairs of distinct content words (sentence IDF ≥ 3, known to the vectors; at least 4 of them)	+
C	c1 sentence cohesion	mean cosine over all pairs of SIF sentence vectors (two halves of the text, of at least 8 tokens, when it has fewer than two embeddable sentences)	+
C	c1a adjacent cohesion	mean cosine of consecutive embeddable sentence vectors	+
C	c2 lexical cohesion	mean over all pairs of sentences of the IDF mass of their shared content words, over the smaller sentence’s IDF mass	+
C	c3 topic drift	Jensen–Shannon divergence between the content-word distributions of the two halves (at least 4 content words)	−
I	loop8	share of 8-byte windows seen earlier in the text	−
I	rep	share of sentences that repeat an earlier sentence	−

Axis	Statistic	Definition	Side
I	d1–d4	distinct-1 to distinct-4 over word n-grams	±
I	i1 verbatim copy	share of tokens inside a word 8-gram that occurs in the pack	±
R	r1 prompt cosine	cosine of the SIF vectors of prompt and text	+
R	r4 key-term coverage	share of the prompt's content words that the text uses, each weighted by its surprisal `−log p(w)` under the pack's unigram model	+
R	r5 informativeness	share of the text's content tokens not in the prompt	±
R	i4 prompt echo	share of the text's word 4-grams that occur in the prompt	−

Also computed and reported, but not scored: raw word-trigram and skeleton log-probabilities (s3, s2), the worst sentence's log-probability, sentence lengths, and mean word rarity. Coherence statistics are one-sided in v2: more cohesion and less drift are never penalised, because cohesion that comes from repetition is caught by the integrity axis.

3.4 From a statistic to evidence

For each statistic `k` and length bucket, let `F` be the mid-rank empirical CDF of the reference values. Inside the 2nd to 98th percentile the z-score is $z = \Phi^{-1}(F(x))$. Beyond it the tail is extended linearly, $z = -2.054 - (q_{0.02} - x)/\sigma_lo$ below and $z = 2.054 + (x - q_{0.98})/\sigma_hi$ above, with $\sigma_lo = (q_{0.16} - q_{0.02})/1.059$ and $\sigma_hi = (q_{0.98} - q_{0.84})/1.059$, floored at a small per-statistic minimum (0.15 for s4, so that one failed check of six counts as moderate evidence). The linear tail keeps the distance of values far outside the human range, where an ECDF would saturate.

The bad-side magnitude is `u = max(0, −z)` for side `+`, `max(0, z)` for side `−`, and `|z|` for `±`. The evidence is `e = max(0, u − 1)`: a dead zone of one standard unit forgives the ordinary spread of human text. A statistic that at least 95% of reference samples have, and the text lacks (for example, no word is known to the vectors), counts as `u = 4`.

3.5 Combination

The evidence of each axis is summed: `L`, `S`, `C`, `I` and `R`. The deviance is

```
D = 30·sat(L − θ_L) + 30·sat(S − θ_S) + 3·max(0, I − θ_I) + C + w_R·R
sat(x) = 1 − exp(−max(0, x)/4)
SCRIBE = exp(−D/5)
```

where `θ_L`, `θ_S` and `θ_I` are the 99th percentiles of `L`, `S` and `I` over the reference samples of the bucket, and `w_R` is 0 in the open-continuation profile and 8 in the prompted profile (whose relevance dead zone is 0.5 instead of 1).

The structure encodes three statements, each tested in Section 4:

- **Form failures are categorical.** A text is either made of words (L) or not, and either in an order people write (S) or not. Human text almost never fails either way, so each counts only beyond what 99% of human samples show, and each saturates: two texts that are both word salad are not told apart by how badly they fail syntax.
- **Degeneration is the worst failure.** Repetition (I) counts beyond the human 99th percentile, with weight 3, and does not saturate.

- **Coherence is graded.** Human text varies continuously in cohesion and less cohesive is worse at every level, so coherence counts from the dead zone on, with no threshold. Inside the word-salad class, coherence orders the texts.

3.6 Profiles, long texts and short texts

The **open-continuation** profile (`v2`) scores a text on its own. The **prompted** profile (`v2p`) adds the relevance axis, read against human prompt–continuation pairs of the genre. Texts longer than the reference samples are cut into consecutive windows of whole sentences (closing a window before 600 bytes once it holds 300) and score the mean over windows. For corpora whose units are short whole texts, such as five-sentence ROC stories, whole reference documents serve as samples.

3.7 Implementation

SCRIBE is a header-only C++20 library using only the standard library, with a command-line tool for document cleaning, prompt extraction, sampling, preparation of model output, control ladders and scoring. Building the Wikipedia pack (32 MiB) and its reference takes under 10 s; the PPMI-SVD vectors take about a minute, once. Scoring needs about 400 MB of memory and runs at roughly a thousand texts per second on one core (Section 6.8). The tool reproduces the earlier v1 ladder to the printed digit.

4. How the design was found

SCRIBE was built in three versions, each tested on a control ladder before anything else. The rule throughout: a design choice is made on development ladders only; a frozen version is then read once on fresh text.

4.1 v0 and v1: percentiles, geometric means and gates

The PLAN that defined SCRIBE converted every statistic to a percentile among human samples (one-sided `min(1, 2F)` or two-sided `1 − |2F − 1|`), took the geometric mean of the sub-scores of each axis and of the axes, and capped a sample at its syntax score when that fell under 0.2 (“word salad cannot be rescued by other axes”). v0 implemented this with a lexicon, a word trigram, bag-of-words cohesion and diversity statistics. On its ladder, which also had a rung of samples from a byte-level count language model, it passed 1 of 6 pairs. v1 added PPMI-SVD sentence vectors, all-pairs cohesion, an integrity gate and length parity, and passed 3 of 6. On the fresh control ladder of this paper it passes 2 of 5 (Section 6.1). Its failures pointed at the combination rather than the statistics: the syntax cap threw away the coherence information that separates shuffled words from random words, and a geometric mean of about eight sub-scores, each roughly uniform on human text, let one weak sub-score pull a human sample down.

4.2 v2: from percentiles to evidence

v2 replaced percentiles by the robust z-scores of Section 3.4, so that a value far outside the human range keeps its distance. The search that followed used dev ladder A (enwik8 [56, 64) MiB) for choices and dev ladder B ([64, 72) MiB) as a check, never the fresh region. Its steps, each measured:

1. **Sum of squared excess z over all statistics, coherence two-sided:** 2 of 5 pairs on ladder A. One unbalanced quote (a single failed well-formedness check) placed a human sample at $z \approx -5$.
2. **Coherence one-sided.** The all-pairs sentence cosine alone separates human from shuffled sentences at 0.989 when only low cohesion counts, against 0.972 two-sided.
3. **No additive weighting passes.** Ordering “shuffled sentences > shuffled words” needs syntax weighted above coherence; but that also amplifies the syntax noise of human samples and breaks “human > shuffled sentences”. Across several thousand additive settings (squared or linear evidence; sum, mean or maximum within an axis; axis weights from 1 to 4), none passed all five pairs on ladder A; the best minimum pair accuracy was 0.926.

4. **Categorical form and integrity.** Counting syntax and integrity only beyond a high human quantile, and coherence from the dead zone on, gave 4 of 5 pairs. The remaining conflict was inside word salad: how badly a text fails syntax varied more than the coherence signal that orders shuffled words above random words.
5. **Context gain.** Raw n-gram probability is low for word salad and also for human text about rare topics. Subtracting each word's frequency-only log-probability leaves the part the context explains; this is SLOR (Section 2) averaged per word. Shuffled words moved from a median bad-side z of 3.1–3.5 (raw probability) to 6.8–8.8 (gain), with none inside the human 99.5th percentile. Samples from a byte-level count language model moved from 19% to 96% beyond the human 99th percentile.
6. **Two form failures.** Once syntax saturates, real words and non-words received the same penalty. Splitting lexical validity (L) from syntax (S), each saturating on its own, passed all five pairs on both development ladders.

The parameters of the final structure are not a knife-edge. Of 2,592 settings of the thresholds, cap, softness, integrity weight and dead zone tried on both development ladders, 1,729 pass all five pairs on both, and every one-step change of a frozen parameter passes. Section 6.7 repeats the main changes on the fresh region. The design was then frozen and committed before the fresh region was read (repository commit `c7fd176`).

The prompted profile's relevance axis was designed afterwards, on development prompts from [56, 64) MiB, against a prompted ladder: true continuation > another document's true continuation > half echo of the prompt > the prompt echoed. Relevance graded from z = 0 orders it best, but then penalises half of all human continuations; a dead zone of 0.5 with weight 8 keeps accuracies of 0.963, 0.967 and 1.000 on that ladder.

5. Evaluation protocol

All splits were fixed before v2 was designed and are listed in the study plan in the repository (`docs/SCRIBE_PAPER_PLAN.md`), which also holds the twenty predictions registered before the first v2 run (Section 6.9).

5.1 Data

Table 4. Genres, packs, references and test texts. MiB = 2^{20} bytes; enwik8 and enwik9 are the first 10^8 and 10^9 bytes of the March 2006 English Wikipedia dump (Mahoney, 2011).

Genre	Pack	Reference	Development	Test
Wikipedia	enwik8 [16, 48) MiB, 2,600 articles, 17.5 MB of prose	[48, 56) MiB, 1,909 + 1,729 samples	ladders A [56, 64) and B [64, 72) MiB	fresh ladder [72, 95.37) MiB; model-ladder prompts from the same range
Web	WebText train, 14,936 documents, 32.2 MB of prose	WebText valid, 5,000 documents	none (no design choice)	WebText test and the GPT-2 output dataset (Radford et al., 2019; Solaiman et al., 2019); the MAUVE human evaluation
Books	80 Project Gutenberg novels, 53.5 MB	20 novels	none	10 novels (split by book)
Stories	WritingPrompts train (Fan et al., 2018), 14,068 stories	WritingPrompts valid	none	HANNA and OpenMEVA-WP stories (prompts from the WritingPrompts test split)
ROC	OpenMEVA ROC train (Mostafazadeh et al., 2016), 87,175 stories	OpenMEVA ROC dev, 4,782 stories (first sentence as prompt)	none	OpenMEVA-ROC stories

Every pack and reference is disjoint from every test text; documents identical to a test text were removed from the packs. The n-gram generators of Section 5.2 train on enwik8 [8, 16) MiB, apart from every SCRIBE pack, and their prompts come from the fresh region. The question-answering test uses Dolly 15k (Conover et al., 2023): two thirds of the rows form the pack, and the rest alternate between reference and test.

5.2 Tests

1. **Control ladders.** Rungs, best first: held-out human samples; sentences shuffled across documents; words shuffled within sentences; random lexicon words (frequency-weighted, in place); random letters (same lengths); the first six words repeated as a sentence. Every control keeps the sample's sentence lengths, terminal punctuation and capitals. 500 standard human samples per ladder. A pair passes at ordering accuracy ≥ 0.95 over all (upper, lower) pairs, ties counting one half (this accuracy is the AUROC), and disjoint 95% bootstrap intervals of the two rung means (1,000 resamples). Run on the fresh Wikipedia region (read once), WebText and books, and on WebText against the Wikipedia reference.
2. **Held-out corruptions.** Nine families, none used in the design: random-word substitution, adjacent word swaps, sentence reordering within the text, sentence repetition, content-word swaps from other documents, typos, truncated sentences, off-topic sentences and dropped words, each at strengths 0, 10, 25, 50, 75 and 100% on 200 fresh human samples (10,800 texts).
3. **Model ladders**, whose order is known in advance. Word n-gram generators (Kneser–Ney, orders 1–5) continue 250 openings of 100–300 bytes from the fresh Wikipedia region; the human rung is the true continuation of the same openings. The GPT-2 output dataset (Radford et al., 2019) has four model sizes from 117M to 1.5B parameters, each under pure sampling and top-k 40, with 1,000 samples each and 1,000 WebText texts. A planned ladder of Pythia training checkpoints and sizes (Biderman et al., 2023) and of GPT-2 decoding settings was not run. Generating it on the available CPU would have taken about 20 hours.
4. **Question answering.** 600 Dolly test questions, each with four answers, best first: the human answer; the human answer to another question of the same category; the question echoed to the answer's length; and the human answer with its words shuffled inside each sentence. Scored with the prompted profile against a Dolly pack and reference.
5. **Human judgments.** The MAUVE human evaluation (Pillutla et al., 2021): 3,240 pairwise judgments of WebText continuations from human writers and eight GPT-2 configurations, on three questions (more interesting, makes more sense, more likely written by a human). HANNA (Chhun et al., 2022): 1,056 stories from ten systems and humans, three raters per story on six criteria, shipped with 72 automatic metrics and five LLM judges. OpenMEVA MANS (Guan et al., 2021): 1,000 ROC and 1,000 WritingPrompts stories rated for overall quality.
6. **Attacks** on 250 fresh Wikipedia prompts: copied pack text, the prompt echoed, keyword stuffing, a fluent generic paragraph with and without the prompt's topic words inserted, randomised numbers, inserted negations, a function-word loop, and off-prompt human text.
7. **Stability:** correlation with length; five seeds; reference size from 50 to 1,000 samples (three draws each); pack size from 8 to 128 MiB.
8. **Ablations** of every axis and every statistic, and of each design decision.
9. **Cost:** time and memory per component and per metric, on one 4-core CPU without a GPU.

5.3 Baselines

All baselines score exactly the texts SCRIBE scores, or a fixed subset of them stated with each result. **Earlier versions:** surface S and SCRIBE v1. **Statistical:** distinct-1–4, repetition rates rep-2–4 and seq-rep-4 (Welleck et al., 2020; for a single text rep-n = 1 − distinct-n, and seq-rep-4 equals rep-4), their product as a diversity score, type–token ratio and gzip compression ratio; per system, self-BLEU and the Zipf coefficient. **Neural:** GPT-2 (124M) perplexity (`openai-community/gpt2`); grammatical acceptability, the mean probability of "acceptable" over sentences from a RoBERTa-base CoLA classifier (`textattack/roberta-base-CoLA`; Warstadt et al., 2019); sentence-embedding coherence, the mean cosine of adjacent sentences under all-MiniLM-L6-v2 (Reimers

and Gurevych, 2019); the RoBERTa-base GPT-2 output detector's probability of human text (`openai-community/roberta-base-openai-detector`; Solaiman et al., 2019); and set-level MAUVE (Pillutla et al., 2021) with GPT-2 features. **Shipped with HANNA:** 72 automatic metrics (among them BLEU, ROUGE, METEOR, chrF, BERTScore, MoverScore, BaryScore, BARTScore and SUPERT) and the ratings of LLM judges (ChatGPT, Mistral-7B, Llama-13B, Beluga-13B). **Published:** the MAUVE scores of the MAUVE human evaluation. Baselines run with their published defaults; nothing is tuned for any metric on test data. The neural baselines are slow on a 4-core CPU, so they read only the first 256 GPT-2 tokens of each text and run on fixed subsets. GPT-2 large, conditional perplexity and an LLM judge of our own were planned but not run.

5.4 Statistics

Rung and system means carry 95% percentile bootstrap intervals (Efron and Tibshirani, 1993; 1,000 resamples). Ordering accuracy counts ties as one half. Agreement with pairwise human choices carries Wilson intervals (Wilson, 1927); Bradley–Terry strengths are fitted to the human choices with a small ridge (Bradley and Terry, 1952). Correlations with ratings are Pearson, Spearman and Kendall's tau-b (Kendall, 1945) with item bootstrap intervals; differences between metrics use a paired bootstrap test with Benjamini–Hochberg adjustment (Benjamini and Hochberg, 1995). The statistics module reproduces the MAUVE paper's published system-level correlations exactly (Spearman 0.810, 0.857 and 0.952 for the three questions) when given the published MAUVE scores.

6. Results

Every number in this section comes from frozen SCRIBE v2. The fresh region was read once, after the freeze. Tables give means with 95% bootstrap intervals where space allows. "Accuracy" is the share of (upper, lower) pairs ordered correctly, ties counting one half (the AUROC). The full tables, with intervals for every cell, are in the results bundle (Reproducibility).

6.1 Control ladders

Table 2. Fresh Wikipedia control ladder: accuracy for each adjacent pair. SCRIBE v2, SCRIBE v1 and surface S use 500 texts per rung; the other metrics use the first 150 texts of each rung. Bold: the pair passes (accuracy ≥ 0.95 and disjoint intervals in the right order).

Metric	human > sentences shuffled	> words shuffled	> random words	> random letters	> repeated phrase	Passed
SCRIBE v2	**0.975**	**0.999**	**0.998**	**0.998**	**1.000**	**5 of 5**
SCRIBE v1	0.888	0.806	0.696	**1.000**	**0.999**	2 of 5
surface S	0.627	0.341	0.750	**1.000**	0.559	1 of 5
GPT-2 perplexity	0.866	**1.000**	0.948	0.000	0.000	1 of 5
CoLA acceptability	0.527	**1.000**	0.715	0.000	0.556	1 of 5
MiniLM coherence	**0.992**	0.007	**0.956**	0.000	0.020	2 of 5
GPT-2 output detector	0.654	0.520	0.943	0.163	0.895	0 of 5
distinct-2	0.273	0.342	0.410	0.360	**1.000**	1 of 5
seq-rep-4	0.463	0.480	0.500	0.500	**1.000**	1 of 5
type–token ratio	0.258	0.742	0.123	0.000	**1.000**	1 of 5
gzip ratio	0.214	0.693	0.226	0.060	**1.000**	1 of 5

SCRIBE v2 is the only metric that passes every pair. Each baseline sees one kind of damage and misses the rest:

- **GPT-2 perplexity** sees word order. But it rates one phrase repeated (perplexity 1.7) far above human text (36.5), and random letters (515) above random dictionary words (2,450), because random letters split into many short, cheap tokens.
- **CoLA acceptability** sees broken syntax inside a sentence. It cannot see shuffled sentences (0.527), and it rates random letters (0.495) above random words (0.057).
- **Embedding coherence** sees shuffled sentences (0.992). It rates a repeated phrase as the most coherent text of all (0.985 against 0.391 for human text).
- **The GPT-2 output detector** passes no pair. It was trained to tell GPT-2 samples from web text, not prose from broken text, and it rates words shuffled within sentences (0.710) as more human than sentences shuffled across documents (0.695).
- **Diversity and compression statistics** see only repetition.

Set-level MAUVE with GPT-2 small features (one set of 100 texts per rung, against 100 other human texts; 50 for the human rung) orders the first four rungs: human 0.994, sentences shuffled 0.979, words shuffled 0.006, random words 0.005 and random letters 0.005. It puts the repeated phrase (0.036) above all three word-salad rungs. Below shuffled sentences its values are all near zero, so it cannot grade broken text, and as a set-level measure it cannot score a single text.

Table 3. SCRIBE v2 on control ladders in other genres, each with its own pack and reference (last row: web text scored against the Wikipedia pack and reference). 500 texts per rung.

Ladder	human > sentences shuffled	> words shuffled	> random words	> random letters	> repeated phrase	Passed	Human mean	Surface S passed
Wikipedia (fresh)	**0.975**	**0.999**	**0.998**	**0.998**	**1.000**	5 of 5	0.916	1 of 5
WebText	**0.966**	**1.000**	**0.985**	**0.999**	**1.000**	5 of 5	0.923	1 of 5
Books	0.785	**1.000**	0.894	**1.000**	**1.000**	3 of 5	0.906	1 of 5
WebText, Wikipedia reference	0.879	**1.000**	**0.997**	**0.998**	**1.000**	4 of 5	0.821	1 of 5

On novels SCRIBE v2 fails two pairs. Sentences shuffled across novels keep a mean score of 0.747 (0.354 on Wikipedia): fiction shares much of its vocabulary across books, so the cohesion statistics that catch shuffled sentences are weaker. The same weakness lets shuffled words (0.003) and random words (0.002) come close. Scored against the wrong genre, human web text keeps 89% of its score, and the ladder loses only its first pair.

6.2 Held-out corruptions

Table 4. Dose–response on 200 fresh human samples: SCRIBE v2's mean at each strength, and Spearman's correlation between strength and score (over the 1,200 texts of a family, and over the six means). Last two columns: the same correlation over texts for v1 and surface S. −1 means the score falls perfectly with damage.

Family	0%	10%	25%	50%	75%	100%	Spearman, texts	Spearman, means	v1, texts	surface S, texts
word substitution	0.905	0.770	0.111	0.004	0.001	0.001	−0.939	−1.000	−0.817	−0.347
typos	0.905	0.861	0.417	0.018	0.001	0.000	−0.937	−1.000	−0.909	−0.970
sentence repetition	0.905	0.887	0.310	0.009	0.005	0.000	−0.917	−1.000	−0.898	−0.903
adjacent word swaps	0.905	0.620	0.067	0.005	0.003	0.003	−0.902	−1.000	−0.677	+0.025

Family	0%	10%	25%	50%	75%	100%	Spearman, texts	Spearman, means	v1, texts	surface S, texts
dropped words	0.905	0.842	0.447	0.195	0.053	0.013	−0.858	−1.000	−0.293	−0.106
content-word swaps	0.905	0.872	0.795	0.534	0.191	0.054	−0.819	−1.000	−0.707	−0.298
off-topic sentences	0.905	0.893	0.737	0.484	0.356	0.364	−0.754	−0.943	−0.592	−0.187
truncated sentences	0.905	0.904	0.899	0.864	0.857	0.922	−0.050	−0.143	−0.060	+0.003
sentence reordering	0.905	0.906	0.905	0.905	0.907	0.906	+0.013	+0.429	−0.002	+0.000

SCRIBE v2 follows seven of the nine families. It falls more steeply than v1 on all seven, and more steeply than surface S on six; surface S, whose vocabulary term counts misspellings directly, falls faster with typos. Off-topic sentences level off at about 0.36: text made entirely of human sentences from other documents is fluent, and only its cohesion is lower. Two families are blind spots. Reordering a text's own sentences changes no statistic, because every cohesion statistic ignores order. Cutting sentences in half and closing them with a full stop leaves them well formed and fluent word by word. At full strength the shorter, simpler text even scores a little higher. The swap generator first swapped overlapping word pairs, which at 100% rotates each sentence by one word instead of scrambling it; it was corrected to disjoint pairs, all results use the corrected version, and the scores of both versions are kept with the study's outputs.

On the first 20 source samples (1,080 texts), where every baseline was run, the baselines follow fewer families (Spearman of the means ≤ −0.9). GPT-2 perplexity follows 6 of 9, CoLA 5, MiniLM coherence 4, the GPT-2 output detector 4, surface S 2, and each count statistic 1 (repetition only); SCRIBE v2 follows 7 of 9 on the same subset. The failures are complementary. GPT-2 perplexity follows truncated sentences (−0.943) and nearly follows reordering (−0.870), the two families SCRIBE cannot see, but it rewards repetition (+1.000), as Holtzman et al. (2020) found. The count statistics rate substituted, dropped and misspelled words as more diverse, and so as better.

6.3 Model ladders

Table 5. Word n-gram generators on fresh Wikipedia prompts (about 250 continuations each). Accuracy is against the next rung down.

Generator	SCRIBE v2	Accuracy vs next	SCRIBE v1	surface S
human continuation	0.911	0.994	0.522	0.911
5-gram	0.089	0.501	0.075	0.873
4-gram	0.081	0.476	0.076	0.870
3-gram	0.081	0.667	0.080	0.872
2-gram	0.051	0.993	0.078	0.871
1-gram	0.001		0.019	0.866

Every n-gram order scores far below human text and below sentences shuffled across documents (0.354), and unigrams are far below bigrams. Over the six systems, Spearman's correlation with the expected order is 0.943 (SCRIBE v1 0.486, surface S 0.829). Orders 3, 4 and 5 are not told apart. All three write locally fluent word salad, and once a text is clearly not prose, SCRIBE's saturating form axes stop measuring how far from prose it is. That is the intended behaviour of the design (Section 3.5), and it means SCRIBE v2 cannot rank models that all fail. Surface S puts the 5-gram model within 0.04 of human text.

Table 6. The GPT-2 output dataset (1,000 texts per system, WebText pack and reference). The two-sided column is an exploratory variant (Section 7), not the frozen metric. GPT-2 perplexity (under GPT-2 small, first 256 tokens),

self-BLEU (lower is more diverse) and the Zipf coefficient use the first 200 texts of each system.

System	SCRIBE v2	two-sided variant	SCRIBE v1	surface S	GPT-2 perplexity	self-BLEU	Zipf
WebText (human)	0.925	0.855	0.534	0.931	35.8	0.085	0.867
xl-1542M, pure sampling	0.826	0.807	0.479	0.929	74.7	0.074	0.859
large-762M, pure sampling	0.816	0.800	0.471	0.927	78.0	0.079	0.866
medium-345M, pure sampling	0.706	0.723	0.364	0.904	176.7	0.058	0.832
small-117M, pure sampling	0.767	0.765	0.426	0.915	103.6	0.067	0.856
xl-1542M, top-k 40	0.969	0.819	0.540	0.929	18.2	0.118	0.921
large-762M, top-k 40	0.968	0.811	0.523	0.927	16.9	0.118	0.918
medium-345M, top-k 40	0.972	0.811	0.549	0.936	19.2	0.115	0.911
small-117M, top-k 40	0.970	0.746	0.478	0.922	14.9	0.141	0.968

Under pure sampling, SCRIBE v2 ranks XL above large above the two small models, but puts medium below small. Under top-k 40 sampling, all four sizes score 0.968–0.972, above WebText itself, and are not ordered. Truncated sampling removes the rare, surprising words that make human text less predictable, and SCRIBE v2 counts only the bad side of each statistic, so text that is more predictable and more cohesive than human text is never penalised. Over the five systems of each decoding family, WebText included, SCRIBE v2's Spearman with the expected order (human, then larger models first) is 0.900 under pure sampling and −0.300 under top-k 40. The count statistics (distinct-n, rep-n, type–token ratio, gzip ratio) run backwards under pure sampling (−0.7 to −0.9), because smaller models write more varied, noisier text, and reach 0.6–0.7 under top-k 40. The exploratory two-sided variant orders the top-k family perfectly (1.000), but it was built after these results were seen. Section 7 discusses this blind spot.

GPT-2 perplexity orders pure sampling as well as SCRIBE v2 does (Spearman 0.900, again with medium below small), and runs backwards under top-k 40 (−0.700), where every model is more predictable than WebText (perplexity 15–19 against 36). It is computed with GPT-2 small, one of the systems, so it favours that system's own samples. Under top-k 40, self-BLEU and the Zipf coefficient put every model on the repetitive side of WebText, the right direction for a quality metric, but neither says anything about size.

6.4 Question answering and attacks

Table 7. Dolly question answering (600 questions, four answers each). Accuracy of the correct answer against each wrong one.

Profile	correct	another question's answer	question echoed	words shuffled	correct > other	correct > echo	correct > shuffled
open (v2)	0.894	0.875	0.372	0.135	0.519	0.813	0.937
prompted (v2p)	0.431	0.005	0.006	0.058	**0.961**	**0.957**	0.863

The open profile cannot see which question an answer belongs to, as intended. The prompted profile separates the right answer from a fluent answer to another question (0.961) and from the echoed question (0.957). It separates right-topic word salad less well (0.863), because many Dolly answers are lists and fragments whose form varies more than prose. Its scale is lower: relevance beyond its small dead zone is common even for correct answers, so only the order of v2p scores is meaningful.

Table 8. Attacks on 250 fresh Wikipedia prompts. Mean score, and the share of prompts where the true continuation beats the attack (ties one half).

Attack	v2 mean	v2p mean	v2: human wins	v2p: human wins
true continuation	0.911	0.481		
copied pack text	0.102	0.001	0.996	0.996
copied pack text, copy statistic i1 off	0.950		0.396	
prompt echoed	0.000	0.000	1.000	1.000
keyword stuffing	0.021	0.000	0.996	1.000
function-word loop	0.000	0.000	1.000	1.000
generic paragraph	0.835	0.012	0.788	0.948
generic paragraph with the prompt's topic words	0.910	0.130	0.662	0.800
another document's continuation	0.913	0.010	0.506	0.980
randomised numbers	0.911	0.484	0.502	0.498
inserted negations	0.907	0.482	0.508	0.502

Repetition, echo, stuffing and copying are caught. Without the copy statistic i1, text copied from the pack scores above the true continuation, so i1 is necessary whenever a model could have seen the pack. A fluent, contentless paragraph is the strongest attack: the open profile rates it near human, and the prompted profile catches it only when it does not borrow the prompt's words (0.948 against 0.800). Wrong numbers and inserted negations are invisible to both profiles: SCRIBE has no notion of truth.

On the first 50 prompts, where the neural baselines were run (unpaired accuracy over all text pairs), GPT-2 perplexity prefers the echoed prompt to the true continuation every time (accuracy 0.000), prefers the generic paragraph in 94% of pairs (0.060), and cannot see copying (0.408). The GPT-2 output detector catches the generic paragraph (0.960) and the function-word loop (0.967), but not copying (0.457). No metric sees wrong numbers or negations reliably: every metric scores between 0.48 and 0.59 on them.

6.5 Agreement with people

Table 9. The MAUVE human evaluation: pairwise judgments of WebText continuations (q1 more interesting, q2 makes more sense, q3 more likely written by a person). Per pair: the share of pairs where the metric prefers the text the person chose. System: Spearman between metric means and Bradley–Terry scores over the eight GPT-2 configurations. The main rows use the 1,667–1,695 judgments with an estimated answer time over 25 seconds, the MAUVE paper's filter, so they compare directly with MAUVE's published correlations. MAUVE is set-level, so it has no per-pair value.

Metric	q1 per pair	q2 per pair	q3 per pair	q1 system	q2 system	q3 system
SCRIBE v2	0.547	0.534	0.541	0.714	0.762	0.857
SCRIBE v2, prompted	0.526	0.529	0.542	0.690	0.786	0.833
SCRIBE v1	0.550	0.541	0.535	0.500	0.500	0.571
surface S	0.534	0.535	0.528	0.476	0.405	0.476

Metric	q1 per pair	q2 per pair	q3 per pair	q1 system	q2 system	q3 system
MAUVE, GPT-2 large features (published)				0.810	0.857	0.952
SCRIBE v2, all 3,240 judgments	0.531	0.527	0.531	0.571	0.810	0.714

People's pairwise choices are hard to predict from one text: no single-text metric here is far above chance (0.53–0.55 per pair). At system level, SCRIBE v2 tracks all three questions better than v1 and surface S, and comes within 0.1 of MAUVE on each (0.714, 0.762 and 0.857 against 0.810, 0.857 and 0.952). MAUVE compares whole distributions in a large model's feature space; SCRIBE averages single-text scores. With eight systems the intervals are wide (q3: 0.21 to 0.88). Without the answer-time filter, SCRIBE v2's system correlations are 0.571, 0.810 and 0.714.

The neural baselines were run on the first 300 pairs only (all judgments; intervals about ±0.06). There, per-pair agreement for q1, q2 and q3 is 0.566, 0.512 and 0.535 for SCRIBE v2; 0.520, 0.540 and 0.533 for GPT-2 perplexity; 0.550, 0.570 and 0.563 for the GPT-2 output detector; 0.570, 0.543 and 0.557 for SCRIBE v1; and 0.43–0.53 for the count statistics. None is clearly better than another at this sample size.

Table 10. HANNA (960 machine-written stories from ten systems; human-written stories excluded). Story-level Kendall τ with the mean human rating of each criterion, and system-level Kendall τ for coherence. RE relevance, CH coherence, EM empathy, SU surprise, EG engagement, CX complexity. Reference-based metrics compare each story with the human story for the same prompt.

Metric	RE	CH	EM	SU	EG	CX	CH, system
SCRIBE v2	0.014	0.194	0.124	0.143	0.168	0.228	0.689
SCRIBE v1	0.074	0.232	0.183	0.209	0.262	0.322	0.600
surface S	0.062	0.215	0.147	0.191	0.229	0.282	0.556
rep-4 (repeated word 4-grams)	0.044	0.194	0.098	0.185	0.199	0.264	0.378
GPT-2 perplexity	−0.047	−0.071	0.000	−0.129	−0.108	−0.171	0.111
BLEU	0.074	0.110	0.133	0.106	0.139	0.189	0.333
ROUGE-L	0.081	0.112	0.145	0.096	0.143	0.168	0.422
METEOR	0.091	0.146	0.181	0.156	0.190	0.277	0.422
chrF	0.096	0.166	0.174	0.171	0.202	0.290	0.467
BERTScore F1	0.132	0.139	0.169	0.123	0.192	0.241	0.556
MoverScore	0.160	0.161	0.149	0.126	0.198	0.250	0.556
BARTScore	0.035	0.034	0.107	0.039	0.078	0.078	0.556
Repetition-3	0.049	0.195	0.107	0.189	0.204	0.260	0.333
Text length	0.114	0.164	0.169	0.198	0.229	0.352	0.333
ChatGPT	0.156	0.202	0.147	0.111	0.181	0.171	0.600
Mistral-7B	0.200	0.261	0.226	0.152	0.240	0.192	0.600
Llama-13B	0.122	0.202	0.185	0.152	0.212	0.253	0.733
Beluga-13B	0.203	0.282	0.259	0.216	0.290	0.304	0.733

On coherence, SCRIBE v2 (0.194) beats BLEU and ROUGE-L (paired bootstrap, $p \le 0.035$ after Benjamini–Hochberg adjustment), matches ChatGPT (0.202) and Llama-13B (0.202), and is below Beluga-13B (0.282, $p = 0.009$). SCRIBE v1 is higher still (0.232), as is surface S (0.215), though neither difference with v2 is significant. A plain repetition statistic, rep-4, ties SCRIBE v2 exactly (0.194): on HANNA, much of what separates coherent from incoherent machine stories is repetition. At system level SCRIBE v2 is far ahead of it (0.689 against 0.378). At system level SCRIBE v2's coherence correlation (0.689) is the highest of all metrics that are not LLM judges,

and below only the two strongest judges (0.733). Relevance to a creative prompt is invisible to it (0.014). When the 96 human stories are included, the reference-based metrics gain sharply (BLEU 0.248 on coherence) because each human story is its own reference, while SCRIBE v2 falls to 0.172.

Table 11. OpenMEVA MANS (1,000 machine-written stories per corpus, overall quality). Story-level Pearson and Kendall, and system-level Kendall over five systems.

Metric	ROC Pearson	ROC Kendall	ROC system	WP Pearson	WP Kendall	WP system
SCRIBE v2	0.189	0.146	0.200	0.057	0.049	0.200
SCRIBE v2, prompted	0.054	0.044	−0.400	0.047	0.040	−0.800
SCRIBE v1	0.108	0.069	0.000	−0.022	−0.007	−0.800
surface S	0.065	−0.005	−0.400	−0.070	−0.045	−0.800
best count statistic	0.109 (distinct-4)	0.086	−0.400	−0.022 (distinct-4)	−0.046	−0.800
GPT-2 perplexity	0.207	0.190	0.000	0.301	0.226	0.600
one rater against the others (human ceiling)	0.848	0.737		0.797	0.709	

On five-sentence ROC stories SCRIBE v2 correlates with overall quality at 0.189, more than twice its predecessors and every count statistic. On WritingPrompts stories the correlation is near zero, and every count statistic is negative. GPT-2 perplexity does better on both corpora: 0.190 on ROC (not significantly above SCRIBE v2, $p = 0.14$) and 0.226 on WritingPrompts ($p = 0.002$). On HANNA, by contrast, it correlates slightly negatively or not at all with every criterion (Table 10). All of these are far below the agreement between OpenMEVA's raters (Kendall 0.71–0.74 for one rater against the mean of the other four). The prompted profile does worse than the open one on both: lexical overlap with a story's opening is not what makes a story good.

6.6 Stability

- **Seeds.** Five further test draws of the fresh ladder all pass 5 of 5 pairs. The weakest pair, human > sentences shuffled, ranges from 0.969 to 0.990, and the human mean from 0.908 to 0.931.
- **Reference size.** With three random reference draws at each size, 100, 200, 500 and 1,000 reference samples per bucket pass 5 of 5 in every draw. With 50 samples every draw fails the first pair (0.885–0.941), and the human mean falls to 0.81–0.85, because the 99th-percentile thresholds and the tail scales are then set by a handful of samples.
- **Pack size.** Packs of 8, 16, 32, 64 and 128 MiB of enwik9, each with its own word vectors, all pass 5 of 5 (first pair 0.971–0.983). A few MiB of text are enough.
- **Length.** On held-out human text, Pearson's correlation of SCRIBE v2 with length in bytes is 0.009 in the standard bucket and 0.041 in the short bucket. Pooled, it is −0.124, because short texts score a little higher on average (0.954 against 0.916).
- **Genre.** Scored against the Wikipedia reference, human web text keeps 89% of its score (Table 3).

6.7 Ablations

Table 12. Fresh Wikipedia control ladder with one part removed or changed (500 texts per rung; italic: the pair fails). Removing any single statistic other than those listed leaves all five pairs passing.

Change	human > sentences shuffled	> words shuffled	> random words	> random letters	> repeated phrase	Passed	Human mean
frozen v2	0.975	0.999	0.998	0.998	1.000	5 of 5	0.916
no lexical axis L	0.977	0.999	0.999	*0.001*	1.000	4 of 5	0.920

Change	human > sentences shuffled	> words shuffled	> random words	> random letters	> repeated phrase	Passed	Human mean
no syntax axis S	0.975	*0.013*	0.999	1.000	1.000	4 of 5	0.918
no coherence axis C	*0.488*	1.000	*0.592*	1.000	1.000	3 of 5	0.982
no integrity axis I	0.987	1.000	0.998	0.998	*0.000*	4 of 5	0.928
no skeleton gain s2g	0.975	0.997	0.992	0.999	1.000	5 of 5	0.915
no context gain s3g	0.975	0.998	0.973	0.991	1.000	5 of 5	0.916
raw log-probabilities instead of gains	0.975	0.961	*0.869*	1.000	1.000	4 of 5	0.916
no saturation of L and S	0.978	0.992	0.994	*0.455*	1.000	4 of 5	0.920
thresholds at the 95th percentile	*0.937*	0.996	0.993	1.000	1.000	4 of 5	0.864
dead zone 0.5	0.977	0.998	0.999	0.998	1.000	5 of 5	0.837
dead zone 1.5	0.969	1.000	0.996	0.998	1.000	5 of 5	0.957
integrity weight 1	0.984	1.000	0.998	0.998	1.000	5 of 5	0.921
plus four diagnostic statistics	0.972	0.999	0.998	1.000	1.000	5 of 5	0.914

Each axis carries one rung, and removing it breaks that rung. Without coherence, shuffled sentences score as high as human text. Without the syntax axis, shuffled words do. Without the lexical axis, random letters tie with real words. Without integrity, a repeated phrase is the best text on the ladder. No single statistic is necessary: the two gains and the well-formedness checks overlap, so either gain alone carries the syntax axis. Replacing the gains by raw log-probabilities, removing the saturation, or lowering the thresholds each breaks a pair, which confirms the three structural decisions of Section 4. The dead zone and the integrity weight can move by half a unit either way without breaking anything; they change only the scale of human scores.

6.8 Cost

Table 13. Time and memory on one Intel Xeon (2.8 GHz, 4 cores), no GPU. SCRIBE uses one core; the Python baselines use two or three threads.

Metric	Texts per second	Threads	Peak memory	One-off setup
SCRIBE v2, Wikipedia pack (32 MiB)	1,054	1	394 MB	pack 3.8 s; vectors 50–60 s once
SCRIBE v2, WebText pack (31 MiB)	726	1	664 MB	pack 7.5 s
count statistics (distinct-n, rep-n, gzip)	2,300–6,100	2		none
MiniLM embedding coherence	35.6	3		model download
CoLA acceptability	5.2	3		model download
GPT-2 perplexity (256 tokens)	4.7	2		model download

Metric	Texts per second	Threads	Peak memory	One-off setup
GPT-2 output detector (RoBERTa-base)	5.9	2		model download
MAUVE features (GPT-2 small, 256 tokens)	10	2–3		model download

Per text, SCRIBE v2 spends about 0.49 ms computing the v1 statistics and 0.20 ms on the v2 additions. Training the word vectors takes 44 s (8 MiB pack) to 99 s (128 MiB) with 377–645 MB of memory, once per pack. SCRIBE is over 200 times faster than GPT-2 perplexity while using half the threads.

6.9 Registered predictions

Table 14. The twenty predictions registered before the first v2 run, scored.

	Prediction (short form)	Result	Verdict
P1	v2 passes all 5 pairs on fresh Wikipedia	5 of 5	hit
P2	surface S passes at most 2 pairs	1 of 5	hit
P3	GPT-2 perplexity fails letters > repeated and ranks the repeated phrase above human text	accuracy 0.000; perplexity 1.7 against 36.5	hit
P4	set-level MAUVE orders the six rungs	4 of 5 adjacent pairs (GPT-2 small features, as amended); the repeated phrase scores above random words and letters	miss
P5	v2 passes ≥ 4 pairs on WebText and on books	5 and 3	miss
P6	Spearman ≤ −0.9 for ≥ 7 of 9 held-out families	7 of 9	hit
P7	Pythia checkpoints in order	not run	untested
P8	Pythia sizes, Spearman ≥ 0.8	not run	untested
P9	v2 orders GPT-2 sizes within one decoding family; MAUVE within both	v2 swaps small and medium (pure) and ties all four (top-k); MAUVE not run	miss
P10	greedy decoding lowest under v2, highest under GPT-2 perplexity	not run	untested
P11	MAUVE human evaluation: q3 system Spearman ≥ 0.6 and per-pair agreement > 0.55	0.857 and 0.541 (all judgments: 0.714 and 0.531)	miss
P12	HANNA coherence: story Kendall ≥ 0.15 and system Kendall ≥ 0.6	0.194 and 0.689	hit
P13	OpenMEVA: story Pearson between 0.1 and 0.4	ROC 0.189, WP 0.057	miss
P14	|Pearson with length| ≤ 0.1 per bucket and pooled	0.009, 0.041; pooled −0.124	miss
P15	web text against the Wikipedia reference keeps < 0.8 of its score	0.89	miss
P16	copied pack text: parity ≥ 0.9 without i1 and < 0.5 with it; stuffing and loop < 0.2	1.04 and 0.11; 0.02 and 0.00	hit
P17	≥ 1,000 texts/s on one core, < 500 MB, ≥ 100 times faster than GPT-2-large features	1,054/s, 394 MB, 224 times faster than GPT-2 small	hit

	Prediction (short form)	Result	Verdict
P18	without C the first pair < 0.95; without s3 the second pair < 0.95	0.488; 0.998	miss
P19	the 1.5B LLM judge fails a ladder pair	judge not run	untested
P20	on HANNA, v2 beats BLEU and ROUGE-L on coherence	0.194 against 0.110 and 0.112	hit

Eight predictions hit, eight missed and four could not be tested. The misses are informative. Books are harder than Wikipedia and web text for SCRIBE's cohesion statistics (P5). SCRIBE's lexical statistics depend less on genre than predicted (P15). Short texts score a little higher than long ones (P14). The syntax axis is more redundant than predicted: the skeleton gain alone carries it (P18). Two predictions about agreement with people were too optimistic (P11, P13). P17 was tested against GPT-2 small, which is cheaper than GPT-2-large featurisation, so the margin against GPT-2 large is larger still.

6.10 Defects found after the freeze

An independent review of the code, done after all results were in, found five defects in the frozen metric and its test generators. None changes a verdict. Fixing them would change v2's definition, so v2 stays as frozen and tested, and the fixes are left for a later version.

1. **Possessive "'s" in the skeleton model.** The skeleton model learns the possessive as "s" but looks it up as "'s", so every possessive is unknown to s2g and the next word loses its context. With the fix, the fresh ladder moves by at most 0.002 (human mean 0.916 to 0.915; human > sentences shuffled 0.975 to 0.973) and still passes 5 of 5.
2. **Curly closing quotes.** The well-formedness checks treat a text that ends in ." as unterminated and cut off, and the word-shuffle control leaves such punctuation on its word. Only the books genre is affected: 16–17% of its samples end that way. With both fixes, the books human mean rises from 0.906 to 0.910, and the ladder still passes 3 of 5.
3. **Short texts in the prompted profile.** The prompted reference holds continuations of 300 bytes or more, so a prompted text under 300 bytes gets no relevance evidence. This affects 26 of the 1,056 HANNA stories, 25 of them from one system, in the prompted profile only.
4. **Adjacent cohesion.** c1a averages over consecutive embeddable sentences, so it skips over a sentence with no known word.
5. **Weak low doses.** At 10% strength the sentence-level corruptions (reordering, repetition, truncation and off-topic sentences) leave most 4–6-sentence samples unchanged, because a tenth of five sentences rounds to none. Reordering stays a no-op for most samples at 25% too. The first step of those dose curves is therefore near zero for every metric. The content-word swap also draws partly from the sample's own words.

The review also found input-handling defects that changed no reported number. An empty pack or reference silently scored every text 1, the prompted profile without its reference silently scored relevance as 0, and one malformed JSON escape could drop a character. These are fixed in the released tool, which still reproduces the published fresh ladder exactly.

7. Discussion

What counts can see. SCRIBE v2's statistics see whether a text is made of words, whether its words follow each other the way they do in human text once their frequency is accounted for, whether its sentences share vocabulary and topic, whether it repeats itself, whether it copies its training text, and how it overlaps with a prompt. With those, it orders every rung of the control ladder on Wikipedia and web text (and 3 of 5 pairs on novels), follows

seven of nine held-out degradations, catches the repetition, echo, stuffing and copying attacks, and puts word-salad generators (n-gram models) far below human text.

What counts cannot see. Four blind spots showed up, each a direct consequence of a design choice:

1. *Sentence order.* All cohesion statistics are symmetric in the order of sentences, so reordering the sentences of a text costs nothing (Table 4). An ordered statistic, such as the entity grid's transition model or the probability of discourse connectives given the previous sentence, is the obvious addition.
2. *Truncation with punctuation.* A sentence cut in half and closed with a full stop is still well formed and still fluent word by word; the context gain ignores the end-of-sentence token. Scoring the end-of-sentence event would catch it.
3. *Truth.* Random numbers and inserted negations keep their human parity. SCRIBE has no belief axis; the PLAN's B axis needs a fact store and claim extraction, which this version does not have.
4. *Too typical.* SCRIBE v2 never penalises text for being more predictable or more cohesive than human text. Truncated sampling (top-k 40) produces exactly such text, and v2 scores it above WebText itself (Section 6.3). An exploratory two-sided variant, which also penalises syntax gains and cohesion that are higher than in human text, puts human text first and orders the four top-k models by size while still passing the control ladder (Table 6; Spearman 1.000 over the five systems, against −0.300 for v2). It does not agree better with people on the MAUVE pairs (per-pair agreement 0.521–0.531, system Spearman 0.38–0.67). Truncated sampling trades diversity for quality (Caccia et al., 2020). Whether a quality metric should penalise excess typicality is a value judgment: detectors exploit it (Gehrmann et al., 2019; Ippolito et al., 2020), people often prefer it, and typical-set decoding (Meister et al., 2023) treats it as a flaw.

Passing the ladder is not the same as ranking fluent texts. v2's dead zones make it a strict judge of whether a text is human-like at all: most human samples score near 1, and every degradation falls far below. Inside the range of fluent text it has little resolution, and on HANNA's graded ratings SCRIBE v1, whose percentile scores keep the ordinary variation of human text, correlates better with people than v2 does. A deployment might use both: v2 as the gate that a model must pass, and a graded within-human-range score to rank models that pass it.

Where the line falls. The control ladder puts "sentences shuffled across documents" just below human text. On Wikipedia every n-gram generator falls far below that rung (means 0.001–0.089 against 0.354). On web text every GPT-2 configuration scores above it (0.71–0.97 against 0.43). So the ladder gives a model builder a concrete first target: a model whose samples do not score above shuffled sentences has not yet learned to write prose, however well it compresses text.

Cost. SCRIBE needs no GPU and no downloaded model. It scores about a thousand texts per second on one core with 0.4 GB of memory. On the same CPU that is about 30 times faster than MiniLM embedding coherence, and about 200 times faster than CoLA, the GPT-2 output detector and GPT-2 small perplexity, each of which used two or three cores (Section 6.8). It can run inside a training loop, on every checkpoint, or on every generated answer.

8. Limitations

- **English only, four genres.** Every pack and reference is English: Wikipedia, web text, novels and short stories. Another language or genre needs its own pack and reference (100 or more reference samples per length bucket and a few MiB of pack text were enough here).
- **Reference dependence.** SCRIBE measures distance from its reference. A reference of the wrong genre lowers human parity (web text against the Wikipedia reference keeps 0.90 of it), and SCRIBE cannot rate text that is better than its reference in ways the statistics do not see.
- **Blind spots.** Sentence order, truncation with punctuation, factual errors and excess typicality (Section 7). The relevance axis is lexical: it separates right from wrong answers on Dolly and on Wikipedia prompts, but not good from bad stories for a creative prompt (HANNA relevance).
- **Graded agreement.** Item-level correlations with people are modest (Kendall up to about 0.2 on stories), like most metrics in Table 10; SCRIBE is stronger at system level and at ordering clear differences than at

ranking fluent texts.

- **Design data.** The control ladder's corruption types were used in design (on development text), so the fresh ladder tests generalisation to new text, not to new kinds of damage. The held-out families, the model ladders and the human judgments test the latter, and they show the limits above.
- **Compute.** Everything ran on one 4-core CPU without a GPU. The neural baselines read only the first 256 GPT-2 tokens of each text and ran on fixed subsets. No LLM judge of our own was run, so the comparison with LLM judges rests on the judges shipped with HANNA. The model ladders stop at GPT-2 XL (1.5B parameters), and the planned Pythia and decoding ladders were not run, so three registered predictions on them are untested.
- **Known defects.** Five small defects found after the freeze are listed in Section 6.10. None changes a verdict, and v2 is reported as frozen.
- **No rating study of our own.** The study plan's own rating test (300 samples rated by the author) was not run; published ratings stand in for it.

9. Conclusion

Counts alone can tell prose from the main kinds of broken text, reliably and cheaply, if two things hold. Each statistic must be read against human text of the same genre. And the evidence must be combined with the right structure: categorical form failures, graded coherence, and a separate penalty for degeneration. The control ladder is a demanding test. None of the standard metrics tried here passes it, and SCRIBE v1, built on almost the same statistics with percentile scores, does not either.

Within fluent text, counts see much less. SCRIBE v2 does not rank GPT-2 sizes. It cannot see order, truth or excess typicality, and it agrees only modestly with people on single texts. Its best use is as a gate and a diagnostic: a cheap, transparent check that a model writes human-like text at all, with per-axis evidence of how it fails. A distributional or learned metric can then rank the models that pass. The clearest next steps follow from the blind spots: an ordered coherence statistic, an end-of-sentence term in the context gain, and a decision on whether text more predictable than people write should count against it.

Reproducibility

Code. All code is in the repository github.com/odin-loki/Cypha; the Zenodo record names the commit. The metric is `native/include/cypha/eval/scribe.hpp`, a header-only C++20 library with no dependency beyond the standard library. The command-line tool is `native/tools/scribe.cpp`. Build it with `cmake -S native -B build && cmake --build build --target scribe`. Its self-test, `scribe selftest`, runs 76 checks and is the CTest `native_scribe_selftest`. Word vectors come from `native/tools/ppmi_svd.cpp` (`g++ -O2 -std=c++20 native/tools/ppmi_svd.cpp -o ppmi_svd`). The evaluation scripts are in `scripts/scribe_eval/`:

- `data.py` downloads and converts every dataset.
- `corrupt.py`, `attacks.py`, `prompted.py` and `qa_ladder.py` build the test texts.
- `baselines.py` computes the baselines, and `meta.py` holds the statistics.
- `report.py` writes every table of Section 6 to `paper/scribe/results/`.

Data. All data is public: enwik8 and enwik9 (Mahoney, 2011), the GPT-2 output dataset, Project Gutenberg, WritingPrompts, HANNA, OpenMEVA, the MAUVE human evaluation, and Dolly 15k. `data.py` writes a manifest with the source, licence, size and SHA-256 hash of every converted file.

Determinism. Every random choice is seeded. The splits and seeds are in the study plan (`docs/SCRIBE_PAPER_PLAN.md`) and the freeze record (`docs/reports/SCRIBE_V2.md`). One control-ladder run (pack, reference and 3,000 test texts) takes about 20 seconds on one core.

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Appendix A. Every parameter of SCRIBE v2

Table A1. Parameters, with the stage at which each was fixed. “Inherited” means it comes from v0 or v1 unchanged; “dev” means it was chosen on development ladders A and B (or, for relevance, development prompts).

Parameter	Value	Fixed
Text cleaning (Wikipedia)	entities, tags, `<ref>`, templates, tables and links stripped; prose lines of ≥ 60 bytes kept	inherited
Text cleaning (other JSONL)	prose lines of ≥ 20 bytes starting like a sentence and ending in . ! or ?	v2
Sentence splitter, tokenizer	abbreviation and initial rules; lowercased words; hyphens split; digit groups kept	inherited
Lexicon	words seen ≥ 2 times in the pack	inherited
Word trigram	interpolated Kneser–Ney, one discount per order `n1/(n1 + 2 n2)`, out-of-vocabulary penalty 8 nats (only in s3, not scored)	inherited

Parameter	Value	Fixed
Skeleton classes	150 most frequent pack words keep their identity; suffix classes -ing, -ed, -ly, nominal (-tion -sion -ment -ness -ity -ism), adjectival (-al -ive -ous -ful -able -ible -ic -ish), -er/-or, plural -s, other alphabetic, non-alphabetic, number	v2
Context gains s2g, s3g	mean over in-lexicon tokens of trigram minus unigram log-probability; s3g needs ≥ 3 such tokens	dev
Content words	sentence IDF ≥ 3 (i.e. in fewer than about 5% of pack sentences)	inherited
Word vectors	PPMI-SVD, window ±5 with 1/distance weights, context smoothing 0.75, 50,000 words, 300 dimensions, `U·S^0.5`, rows centred and normalised	inherited (C1)
Sentence vectors	SIF, `a = 10⁻³`, first principal component of up to 60,000 pack sentences removed (15,680 for the Wikipedia pack)	inherited
Short-text cohesion	a text with fewer than two embeddable sentences is cut into two halves of its tokens (≥ 8 tokens)	inherited
8-gram copy set	all word 8-grams of the pack, hashed to 64 bits	v2
Length buckets	short < 300 bytes ≤ standard; reference samples 60–260 and 400–600 bytes; a bucket with < 20 reference samples falls back to the other	inherited / v2
z-score	mid-rank ECDF inside the 2nd–98th percentile; linear tails with scales from the 2nd–16th and 84th–98th percentiles	dev
Tail floors	s1 0.01; s2g, s3g 0.05; s4 0.15; c0, c2 0.005; c1, c1a, c3 0.01; loop8 0.01; rep 0.05; d1, d2 0.01; d3, d4 0.005; i1 0.01; r1 0.01; r4, r5 0.02; i4 0.01	dev (s4), otherwise first choice
Missing statistic	counts as `u = 4` if ≥ 95% of reference samples of the bucket have it (not for relevance)	dev
Sides	Table 1; coherence one-sided	dev
Dead zone	1 (relevance 0.5)	dev
Thresholds θ_L, θ_S, θ_I	99th percentile over the reference samples of the bucket	dev
Saturation	height 30, softness 4, for L and S	dev
Weights	integrity 3, coherence 1, relevance 0 (open) or 8 (prompted)	dev
Score	`exp(−D/5)`	dev (monotone; affects no ordering)
Long texts	windows of whole sentences: a window closes before 600 bytes once it holds 300; a last window under 300 bytes joins the previous one; the score is the mean over windows	v2

Exploratory option (not v2). `twosided=1` also penalises s2g, s3g and every coherence statistic when they are higher than human text; it is reported only in Sections 6.3 and 6.4 and was decided after those results were seen.